GME Analysis — Hubbard L=4 Ne=4 t=1.0 U=4.0 — n=3 (mixed)

```
System:  L=4  Ne=4  t=1.0  U=4.0  mode=uniform
RDM:  n=3  sites=(0, 1, 2)  dim=8x8
Tr(rho) = 1.00000000   Tr(rho^2) = 0.46262004   rank = 4

============================================================
  LOWER BOUNDS
============================================================


  C_LB  notes eqs.(6-9)       = 0.00000000
    best z1 = [0, 0, 0]
    best z2 = [1, 1, 1]

  C_LB  best of all               = 0.00000000

============================================================
  CONVEX ROOF
============================================================

  C_num (existing numerical)       = 0.50351431
  C_GME (unitary optimization)     = 0.50351431
    restarts = 200   optimizer = L-BFGS-B

  Decomposition from unitary optimization:
    i=0: p=0.250000  C_GME=0.503514  contrib=0.125879
    i=1: p=0.250000  C_GME=0.503514  contrib=0.125879
    i=2: p=0.250000  C_GME=0.503514  contrib=0.125879
    i=3: p=0.250000  C_GME=0.503514  contrib=0.125879

============================================================
  C_LB=0.000000 <= C_GME=0.503514 <= C_UB=0.912871
```

GME Analysis — Hubbard L=4 Ne=4 t=1.0 U=4.0 — n=4 (pure)

```
System:  L=4  Ne=4  t=1.0  U=4.0  mode=uniform
RDM:  n=4  sites=(0, 1, 2, 3)  dim=16x16
Tr(rho) = 1.00000000   Tr(rho^2) = 1.00000000   rank = 1

  C_GME exact (eq.1) = 0.91287093

=========================================================
  LOWER BOUNDS
=========================================================


  C_LB  notes eqs.(6-9)       = 0.66666667
    best z1 = [0, 1, 0, 1]
    best z2 = [1, 0, 1, 0]

  C_LB  best of all               = 0.66666667
  Gap  C_exact - C_LB_best        = 0.24620426

=========================================================
  CONVEX ROOF  (pure state)
=========================================================

  C_num (existing numerical)      = 0.91287093
  C_GME (unitary optimization)    = 0.91287093
  C_exact (eq.1)                  = 0.91287093

=========================================================
  C_LB=0.666667 <= C_GME=0.912871 <= C_UB=0.912871
  C_exact = 0.912871
```